

Hammad Ahmed

API 571 – Corrosion and Material | API 580 – Risk-Based Inspection | CENOSCO IMS PEI (v1) | Pakistan Engineering Council Accreditation | Metallurgical & Materials Engineer

Year of Birth: 2001

Nationality: Pakistani

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Qualification / Educational Background:

- B.Sc Metallurgy & Materials Engineering
(Institute of Metallurgy & Material Engineering University of the Punjab, Lahore), (2019-2023)
- Pakistan Engineering Council Accreditation

Professional Trainings / Certifications

- API 571 – Corrosion and Material (Cert No.129029)
- API 580 – Risk-Based Inspection (Cert No.125983)
- CENOSCO IMS PEI (v1)
- AMPP – Basics of Cathodic Protection

Publications:

- Hammad Ahmed et al 2025 Eng. Res. Express 7 035435. Effect of tool rotational speed on the mechanical and electrochemical properties of friction stir welded 5083 Al-Mg alloy.

Link: <https://iopscience.iop.org/article/10.1088/2631-8695/ae04c6>

EXPERIENCE

SGS Pakistan Pvt. Ltd. (2024 to Present):



Corrosion Engineer

(Corrosion Engineer-Pakistan: Pipeline, Cathodic Protection, Internal Corrosion, Coatings)
(Corrosion Engineer-Middle East Region: Pipeline. Cathodic Protection, Internal Corrosion, Coatings)

Summary of Experience:

Over the past +1 years, after completing my degree in Metallurgy & Materials Engineering, I have successfully contributed to more than 40+ projects related to Onshore Pipeline Integrity Management, Internal Corrosion Projects and Cathodic Protection Systems, including EPC and turnkey solutions covering design, procurement, installation, and commissioning. My experience also includes CP audits, troubleshooting, system rehabilitation, coating inspections, internal corrosion solutions, and corrosion monitoring assessments using coupons, probes, and data loggers. These projects have been executed for clients across various sectors such as Oil & Gas, Power, Fertilizers, and Chemicals within Pakistan & International.

I'm responsible to handle projects assignments as:

- Pipeline Integrity Management (Oil and Gas Up-Stream, Mid-Stream & Down-Stream Pipelines),
- Cathodic Protection Systems EPC - Turnkey Solutions – (Design, Procurement, Installation & Commissioning), CP Auditing Troubleshooting Rehabilitations & Consultancy,
- Develop complex technical CP designs including calculations, material specification, QA/QC documents & engineering drawings
- Internal Corrosions Solutions Corrosion Monitoring Assessments Analysis (coupons, probes & data loggers) Preventions Mitigations,
- Integrity Assessments particularly NACE ECDA-External Corrosion Direct Assessment Methodologies, NACE ICDA-Internal Corrosion Direct Assessments Methodologies & NACE SCCDA-Stress Corrosion Cracking Direct Assessment Methodologies,

- Manage site survey teams for CIPS, DCVG, ACVG, ACCA, PCM, Soil Resistivity Surveys, Pipeline Elevation Profile Survey, CP Audits, Underground Pipeline/Cables Locating & Mapping, pH Evaluation (in the ditch examination), Interference /Stray Current Surveys
 - Bell-Holes Management (in the ditch evaluation) NDT Visual Inspections ASME B31G Assessments & decisions for Run-Repair-Replace-Rerate & Pipe/Coating/CP repair reinstate QAQC & backfilling, auditing, trainings, Projects Management.
-

Major Projects:

- As a **Project Engineer** responsible for the Rehabilitation of existing CP System of whole plant, Supply of material, Installation & Testing of **Balloki Power Plant**. (2024 to 2025)
 - As a **Project Engineer** responsible for the CP EPC (transformer rectifier based) ICCP system - detailed engineering design, field surveys, supply, installation, connection, testing, pre-commissioning, and follow-up testing of complete and fully functional impressed current cathodic protection system ICCP, of complete underground pipeline of HRS extension at **PSO terminal Allama Iqbal International Airport Lahore**. (2025 to 2025)
-

Other Training / Course - During Engineering:

2022 (Internship/Training) Descon Technical Institute, Lahore



API INDIVIDUAL CERTIFICATION PROGRAMS



verifies that

Hammad Ahmed

HAS MET THE ESTABLISHED AND PUBLISHED REQUIREMENTS FOR API CERTIFICATION AS AN

API 571 CORROSION AND MATERIALS INSPECTOR

CERTIFICATION NUMBER **129029**

ORIGINAL CERTIFICATION DATE **August 31, 2025**

CURRENT CERTIFICATION DATE **August 31, 2025**

EXPIRATION DATE **August 31, 2028**

DIRECTOR, API INDIVIDUAL CERTIFICATION PROGRAMS



ICP



To verify authenticity, scan the above QR code
or go to <http://inspectorsearch.api.org>

API INDIVIDUAL CERTIFICATION PROGRAMS



verifies that

Hammad Ahmed

HAS MET THE ESTABLISHED AND PUBLISHED REQUIREMENTS FOR API CERTIFICATION AS AN

API 580 RISK-BASED INSPECTION INSPECTOR

CERTIFICATION NUMBER **125983**

ORIGINAL CERTIFICATION DATE **April 30, 2025**

CURRENT CERTIFICATION DATE **April 30, 2025**

EXPIRATION DATE **April 30, 2028**

DIRECTOR, API INDIVIDUAL CERTIFICATION PROGRAMS



To verify authenticity, scan the above QR code
or go to <http://inspectorsearch.api.org>.

Certificate

This certificate acknowledges that

Hammad Ahmed

has successfully completed the course

IMS PEI: Complete (v1)

February 8, 2025

Credit Hours: 32

Fundamentals, Circuits, Degradation Management, IMS Features

Luiz Curado
Rowan Vaduganathan





CERTIFICATE OF PROFESSIONAL DEVELOPMENT HOURS

Be it known

Hammad Ahmed

has attended an AMPP sponsored meeting on

Basics of Cathodic Protection

In recognition whereof, **1.00** Professional Development Hour(s) are awarded.

16 July 2024

DATE

Tasos Kostrivas

Dr. Tasos Kostrivas

CHAIR - AMPP UK CHAPTER

رجسٹریشن نمبر
Registration Number
2019-ucc-69

University of the Punjab



جامعہ پنجاب

رول نمبر
Roll Number
19444E-S1-330

This university hereby confers upon
Hammad Ahmed

son of **Khalid Mehmood**

of the **Institute of Metallurgy and Materials Engineering,
University of the Punjab, Lahore**

یہ جامعہ

حماد احمد

بن **خالد محمود**

از انسٹیٹیوٹ آف میٹالرجی اینڈ میٹیریئلز انجینئرنگ
جامعہ پنجاب، لاہور

the degree of

**Bachelor of Science (Engineering)
Metallurgy & Materials Engineering**

together with all the rights, privileges and
honours belonging to this degree.

Session 2019-2023

C.G.P.A. 3.64

کو بی۔ ایس سی (انجینئرنگ)

میٹالرجی اینڈ میٹیریئلز انجینئرنگ

کی سند عطا کرتی ہے مع تمام حقوق، امتیازات

اور اعزازات جو اس سند کے ساتھ وابستہ ہیں۔

دورانیہ ۲۰۱۹-۲۰۲۳ء

سی۔ جی۔ بی۔ ای۔ ۳.۶۴



محمد بلیمغ الرحمن

Chancellor

چانسلر

Controller of Examinations

کنٹرولر امتحانات

Lahore : Pakistan

لاہور: پاکستان

BSEN 004549

Hammad Ahmed Projects with SGS
Pipeline Integrity Management, Cathodic Protection, Internal Corrosion, Coatings
Projects Profile From 2024-2025

S.#	Project Name (Scope Of Work)	Location	Client Name	Period From	Period To
1	CP EPC (TRANSFORMER RECTIFIER BASED) ICCP SYSTEM - DETAILED ENGINEERING DESIGN, FIELD SURVEYS, SUPPLY, INSTALLATION, CONNECTION, TESTING, PRE-COMMISSIONING, AND FOLLOW-UP TESTING OF COMPLETE AND FULLY FUNCTIONAL IMPRESSED CURRENT CATHODIC PROTECTION SYSTEM ICCP, OF COMPLETE UNDERGROUND PIPELINE OF HRS EXTENSION AT PSO TERMINAL ALLAMA IQBAL INTERNATIONAL AIRPORT LAHORE INCLUDING SOLAR DESIGNING WORKS ONLY	Allama Iqbal International Airport, Lahore	IIW - INDUSTRIAL ENGINEERS & CONTRACTORS PVT. LTD. (Allama Iqbal International Airport, Lahore)	2025	2025
2	Call-Out Agreement (Indirect Inspection Survey as per NACE SP0502) Work Contract 21KM ECDA	Karachi, Pakistan	PAKISTAN PETROLEUM LIMITED	2025	2025
3	CONSULTANCY SERVICES FOR ASSET INTEGRITY GAP ANALYSIS STUDY TENDER NO. PDIGENIPT128749124	Karachi, Pakistan	Pakistan Petroleum Limited	2025	2025
4	Call-Out Agreement (Indirect Inspection Survey as per NACE SP0502)	Karachi, Pakistan	PAKISTAN PETROLEUM LIMITED	2025	2027
5	Monthly audit of CP system at CTS	Karachi, Pakistan	FOTCO +92 334-9244888 Email : sm.usama@fotco.pk"	2024	2024
6	Design , Supply , installation ,testing/commissioning of CP system for new 12" dia, 4.7 km MS pipeline and for in-service 16" dia, 4.7 km HSD pipeline	Karachi, Pakistan	Hascol Petroleum Limited (Mehmood Kot)	2024	2024
7	DCVG & CIPS Survey of Naimat Field	Karachi, Pakistan	United Energy Pakistan	2024	2024
8	Supply and Installation of Surge Arrestor	Karachi, Pakistan	Al-Tariq Constructors (Pvt.) Ltd, 1301 - 1302 Uni Centre, I.I Chundrigar Road, Karachi. 74400 Pakistan	2024	2024
9	to inspect the retrieved pipes of Unarpur flow line (8" x 5 kms) for checking internal and external defects to ensure their integrity	Karachi, Pakistan	UEPL	2024	2024
10	Design, Supply, Installation, Testing and Commissioning of CP Works of Pipeline	Karachi, Pakistan	Al-Tariq Constructors (Pvt.) Ltd, 1301 - 1302 Uni Centre, I.I Chundrigar Road, Karachi. 74400 Pakistan	2024	2024
11	inspection of CP system for first quarter of Y-2024	Karachi, Pakistan	Total Parco, Shikarpur Terminal, Near Parco Pumping Station 3, Kandh Kot Road	2024	2024
12	Inspection for conductor pipe of Tor-1	Karachi, Pakistan	UEP	2024	2024
13	SERVICES REQUIRED FOR DCVG SURVEY OF ONSHORE SPM CRUDE PIPELINE (length: 03 KM)	Karachi, Pakistan	Cnergico Pk Limited	2024	2024
14	Inspection of Retrieved pipeline at MKK Sutiari field	Karachi, Pakistan	UEP	2024	2024
15	Leak Detection Survey for Underground Domestic Water supply line and Fire Water Line	Karachi, Pakistan	HUBCO	2024	2024
16	inspection of CP system for first quarter of Y-2024	Karachi, Pakistan	Total Parco, Shikarpur Terminal, Near Parco Pumping Station 3, Kandh Kot Road M +92 (307) 5504855"	2024	2024
17	Annual Contract for CP Inspection and Audit at PGCL Terminal	Karachi, Pakistan	FOTCO LNG Terminal	2024	2024
18	Inspection of NAKA Retrieved Pipeline at MKK Sutiari Petrocon workshop	Karachi, Pakistan	UEP	2024	2024
19	Combined PCM and DCVG Survey Of POD Pipelines (28 Km.s)	Karachi, Pakistan	UEP	2024	2024
20	Combined PCM and DCVG Survey Of Thebo Sumar Pipeline	Karachi, Pakistan	UEP	2024	2024
21	Inspection for retrieved pipeline from Siraj South-1 to Bijoro-2 laying at Petrocon yard (Tando Adam).	Karachi, Pakistan	UEP Siraj-South Bijoro	2024	2024
22	Integrity assurance of retrieved flow line from Miano	Karachi, Pakistan	UEP	2024	2024
23	External Corrosion Direct Assessment Study of 07 x Production Critical Pipeline loops	Karachi, Pakistan	Mari Petroleum Company Limited, Daharki District Ghotki Sindh, Pakistan www.mpcl.com.pk	2024	2024
24	PCM and DCVG survey of 12" Sutiari-Naimat line (12 Kms)	Karachi, Pakistan	UEP	2024	2024
25	Inspection of CP system on cooling water pipeline at EPTL plant	Karachi, Pakistan	Engro powergen Thar (EPTL Plant)	2024	2024
26	Underground Firefighting Pipeline Leakage Detection	Karachi, Pakistan	Midas Safety, Beltexco Limited Plot No. 8-12, Sector C-VII KEPZ, Karachi, Pakistan.M: +92 300 2649971"	2024	2024
27	INTERNAL CORROSION DIRECT ASSESSMENT STUDY OF 01 X PRODUCTION CRITICAL PIPELINE (20-Inch Pipeline From MD-02 to MD-15 to SGPC)	Karachi, Pakistan	MPCL	2024	2024
28	ECDA survey of GS Carbon steel feeder lines at Gambat South Field Shahdadpur (3 lines of almost 59 Km.s)	Karachi, Pakistan	Pakistan Petroleum Limited	2024	2024
29	Design , Supply , installation ,testing/commissioning of CP system	Karachi, Pakistan	PETROCON (PVT) LTD, ENGINEERS, FABRICATORS & ERECTORS.	2024	2024
30	Rehabilitation of existing CP System of whole plant, Supply of material, Installation & Testing	Karachi, Pakistan	HEI-HRL Joint Venture (Balloki Power Plant) "Hamid Javed hjaved@hei-om.com"	2024	2024
31	Services Quotations & Performance Evaluation of AGL Cathodic Protection System	Karachi, Pakistan	Agritech Ltd (formally PAK-American Fertilizer Ltd) Mianwali-Pakistan	2024	2024
32	DCVG and CIPS Surveys of Cross Country Gas Lines Including Health Assessment of CP System	Karachi, Pakistan	Engro Fertilizers Limited	2024	2024
33	Inspection of CP System for Q3rd Quarter	Karachi, Pakistan	Total Parco Shikarpur Terminal, Near Parco Pumping Station 3, Kandh Kot Road	2024	2024
34	Annual Maintenance Contract for Inspection of CP System CTS	Karachi, Pakistan	FOTCO	2024	2024
35	Annual Maintenance Contract for Inspection of CP System at PGCL	Karachi, Pakistan	FOTCO	2024	2024
36	Health Assessment of Efert Cathodic Protection System	Karachi, Pakistan	Engro Fertilizers Limited	2024	2024
37	ECDA Survey includes DCVG and CIPS Survey at Well SCN X-1 and Kabir X-1 Feederline	Karachi, Pakistan	Pakistan Petroleum Limited	2024	2024
38	DCVG Survey of 14" Dia, 82KM Pipeline (Pipri terminal to Hubco Power Station)	Karachi, Pakistan	Asia Petroleum Limited	2024	2024
39	ECDA Survey Includes DCVG and CIPS Survey at Well Shahdad X-1 Feederline.	Karachi, Pakistan	Pakistan Petroleum Limited	2024	2024
40	Services required to locate 30-inch product pipelines outside terminal with specialized line detector equipment and other equivalent technology	Karachi, Pakistan	Karachi Hydrocarbon Terminal Limited	2024	2024
41	PCM & DCVG survey of Siraj S-Naimat line Service (9 KM)	Karachi, Pakistan	United Energy Pakistan	2024	2024
42	Supervision & Witness charges of Jetty Piles Inspection	Karachi, Pakistan	FOTCO	2024	2024
43	Inspection of CP System for Q4th Quarter	Karachi, Pakistan	Total Parco	2024	2024
44	Demonstration of DCVG and CIP survey of Sample Underground Piping	Karachi, Pakistan	FFBL	2024	2024
45	PCM & DCVG survey of Badin lines Service	Karachi, Pakistan	United Energy Pakistan	2024	2024
46	INSPECTION OF RLNG PIPELINE - 30" - 14KM	Karachi, Pakistan	FOTCO	2024	2024
47	LENGTH (DCVG APPLICABLE AT 9KM	Karachi, Pakistan	k-electric, arsalankhan@ke.com.pk	2024	2024
48	UNDERGROUND PIPELINE Locating Service	Karachi, Pakistan	KHT, mailto:sau@vti.com	2024	2024
49	Monitoring of Sacrificial CP System	Hub, Pakistan	Cnergico Pk Limited	2024	2024
50	INSPECTION SERVICES REQUIRED FOR KHT UNDERGROUND PIPELINE INSPECTION. FROM KHT TO FOTCO 3 PIPELINES (EACH LENGTH 1350M) AND FOTCO TO PAPCO 2 PIPELINES (285M & 255M).	Karachi, Pakistan	FOTCO - FAUJI OIL TERMINAL AND DISTRIBUTION COMPANY LTD	2024	2024

PAPER

Effect of tool rotational speed on the mechanical and electrochemical properties of friction stir welded 5083 Al-Mg alloy

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Effect of tool rotational speed on the mechanical and electrochemical properties of friction stir welded 5083 Al-Mg alloy

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Keywords: friction stir welding, aluminum-magnesium alloys, phase analysis, mechanical analysis, corrosion analysis

Abstract

While friction stir welding (FSW) is a sustainable method for joining aluminum alloys, the effect of rotational speed on the combined microstructural, mechanical, and corrosion behavior of 5083 Al-Mg alloy is not well established. Most studies treat these properties separately, creating a gap in understanding their interdependence. Closing this gap is essential for optimizing FSW parameters for durable marine applications. Therefore, this study investigates the impact of varying rotational speeds (600, 700, and 800 RPM) on the metallurgical evolution, mechanical performance, and electrochemical behavior of friction stir welded 5083 Al-Mg alloy. Welds were produced using a butt joint configuration with a fixed tool traverse speed of 180 mm min⁻¹. Comprehensive analyses were conducted, encompassing x-ray diffraction analysis, mechanical testing, and electrochemical corrosion evaluations. The findings reveal that a rotational speed of 800 RPM delivers the most optimal weld characteristics compared to 600 RPM and 700 RPM. Specifically, the weld fabricated at this speed exhibited the highest tensile strength (342.48 MPa), minimal residual stresses, a highly uniform phase formation, and exceptional corrosion resistance, evidenced by an impressively low corrosion rate of 0.01 mpy.

1. Introduction

Aluminum alloys are highly valued for their exceptional attributes, including an impressive strength-to-weight ratio, outstanding corrosion resistance, and cost-efficiency [1, 2]. Among these, the 5083 aluminum alloy distinguishes itself with superior performance in extreme environments, demonstrating exceptional resistance to seawater and industrial chemicals [3]. Moreover, this alloy retains its mechanical strength even after welding, making it ideal for use in challenging and demanding applications [4]. Primarily alloyed with magnesium and supplemented by minor amounts of manganese, chromium, and silicon, 5083 Al-Mg alloy is a non-heat-treatable, strain-hardened alloy widely used in marine, automotive, aerospace, and structural applications due to its excellent corrosion resistance, high strength-to-weight ratio, and good weldability. In spite of these diverse applications, there is a discrepancy in joining this alloy perfectly [5]. This is because conventional fusion welding often causes defects such as porosity, hot cracking, and loss of mechanical properties in this alloy. This fact highlights the pressing need to develop advanced and optimized joining techniques to maximize its potential and utility [6].

Friction stir welding (FSW), being a solid-state joining technique, avoids melting-related problems and produces defect-free joints with refined microstructures and improved corrosion resistance [7, 8]. This method is particularly well-suited for the 5083 Al-Mg alloy, offering significant advantages over conventional fusion

welding techniques by minimizing microstructural heterogeneity and enhancing both mechanical and electrochemical properties [9]. During FSW, heat is generated as the rotating tool penetrates the interface between the metals and moves along the weld line [10]. This localized heating softens the material without reaching its melting point, enabling the formation of a robust, solid-state joint. The intense heat generated in the weld zone (WZ) promotes the creation of a uniform, recrystallized grain structure, a hallmark of FSW [11]. Surrounding the WZ, the thermo-mechanically affected zone (TMAZ) experiences microstructural modifications driven by the mechanical stirring and deformation effects of the tool pin. The ability of FSW to precisely control these microstructural transformations is fundamental to its effectiveness, yielding joints with exceptional performance characteristics [12].

FSW is widely acknowledged for its ability to produce welded joints with superior mechanical properties compared to conventional fusion welding techniques [13]. This process generates localized heat through the interaction of the rotating tool with the joining metals, leading to the development of distinct weld regions: the WZ, TMAZ, heat-affected zone (HAZ), and the unaffected base metal (BM). Each of these zones exhibits unique microstructural and mechanical attributes, collectively influencing the overall integrity and performance of the welded joint [14].

The performance characteristics of FSW can be precisely tailored by adjusting its fundamental process parameters, allowing for the optimization of weld quality and functionality. The four principal parameters that significantly influence FSW outcomes are rotational speed, traverse speed, tilt angle, and tool geometry [15]. Variations in these parameters markedly affect the weld's microstructure, mechanical properties, and corrosion performance. Liu *et al* [16] highlighted the sensitivity of tensile properties and fracture behavior to rotational speed parameters. At a low rotational speed of 350 RPM, the mechanical properties were inadequate, with tensile strength reaching only 310 MPa and hardness 60 HV, resulting in poor bonding between the metal plates. In contrast, Thomas *et al* [17] reported that an optimal rotational speed of 400 RPM produced superior tensile strength (410 MPa) and hardness (79 HV). However, excessive heat generation at 1250 RPM degraded the tensile strength to 250 MPa, attributed to disruptions in molecular interactions during the FSW welding process. Sung-Hyeon *et al* [18] investigated the FSW of 5083 Al-Mg alloy at 400 RPM, finding a high corrosion rate of 8.45 mpy in the WZ region, which was associated with an irregular and non-uniform microstructure, leading to inadequate corrosion resistance in a 3.5 wt% NaCl solution. Similarly, Behnagh *et al* [19] examined the effects of very high rotational speeds (1400 RPM) and reported excellent mechanical properties but reduced corrosion resistance. This decline was attributed to excessive heat input and intensified stirring action, resulting in the formation of coarse and less corrosion-resistant microstructures. Some researchers have also been studied FSW process on pure copper [20] and dissimilar metals (aluminum alloy and copper) [21]. Ahmed *et al* [22] studied the FSW of rolled plates AA7075-T6 and AA5083-H111 dissimilar butt joint welding. At a constant rotational speed of 300 rpm and traverse speeds of 50–200 mm min⁻¹. The microstructure revealed a grain size reduction increased welding speed resulted grain refinement from 9 to 3 µm in the nugget zone (NG), depending on welding speed. The hardness profile of AA7075 joints exhibited softening in the NG zone, while a tensile strength of 245–267 MPa with 77%–87% joint efficiency. Sun *et al* [23] studied the behavior of solutionized, quenched and naturally aged Al-2024 and annealed Al-5083 alloys were FSW at 1200 rpm and 20–40 mm min⁻¹ in T-joint configuration. It has been observed that with low heat input, material is not plasticized properly, resulting in defects and inadequate dynamic recrystallization (DRX) which causes bad weld morphology. Mechanical experimentation exhibited that UTS of the weld has achieved 86% of Al-2024 base strength, and for the Al-5083 base, it is increased by 18%. Nui *et al* [24] studied the corrosion behavior of FSW Al-2024 and Al-7075 alloy plates in an EXCO solution (4 M NaCl, 0.5 M KNO₃ and 0.1 M HNO₃) via three electrodes system. The rotational and transverse speed was set to 600 rpm and 200 mm min⁻¹ respectively. Corrosion behavior revealed that the open circuit potential (OCP) of the stir zone (SZ) matched that of the BM located on the retreating side (RS). The fine recrystallized grains in the SZs exhibit a lower corrosion resistance than those of BMs. Collectively, these studies emphasize the necessity of a meticulous balance in FSW parameters to optimize both mechanical properties and corrosion resistance, ensuring robust and durable welded joints.

This study investigates the impact of rotational speed on the metallurgical, mechanical properties, and electrochemical behavior of 5083 Al-Mg alloy joints produced by FSW process. Medium-range rotational speeds of 600, 700, and 800 RPM were selected for analysis, maintaining a constant tool traverse speed of 180 mm min⁻¹. This methodology enabled an in-depth assessment of the mechanical performance and corrosion resistance of the welded specimens.

2. Materials and methods

The 5083 Al-Mg alloy, characterized by the chemical composition outlined in table 1, was selected as the base material in plate form with dimensions of 285 × 95 × 6 mm. Prior to welding, the plates underwent meticulous

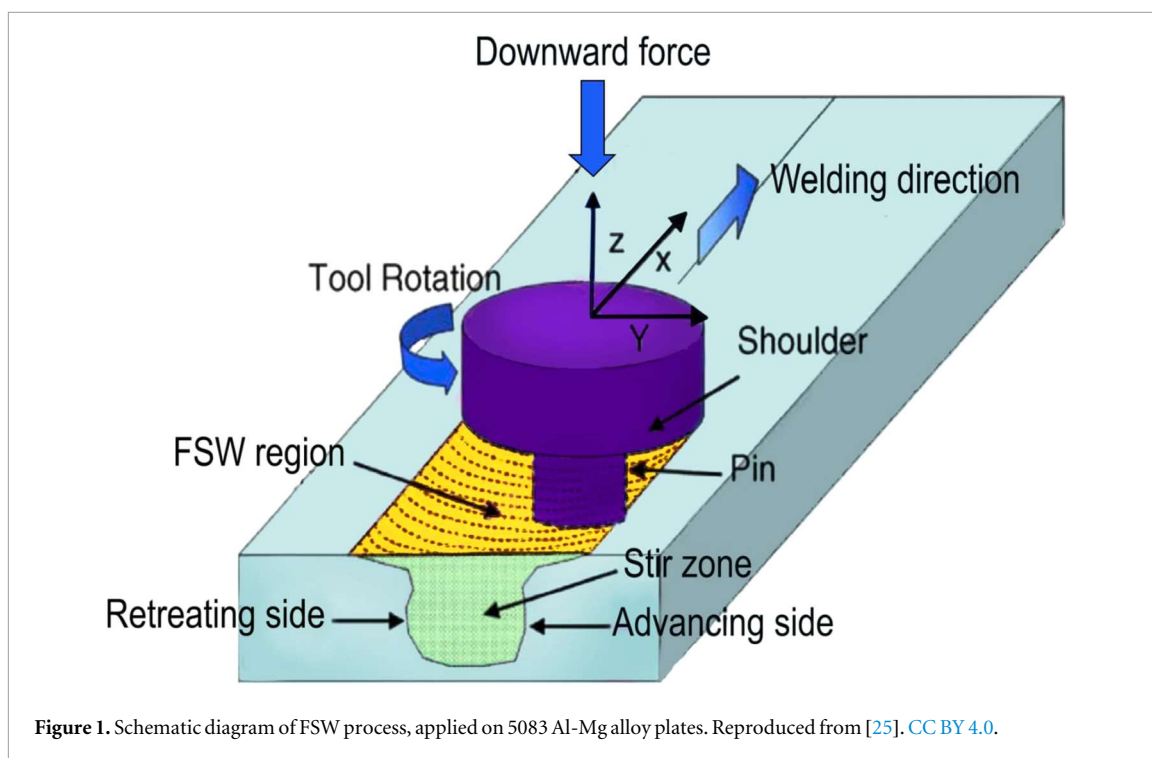


Table 1. Chemical composition of 5083 Al-Mg alloy plates (wt%).

Mg	Mn	Fe	Zn	Si	Cr	Al
4.5	0.73	0.35	0.015	0.15	0.07	Bal.

surface preparation by being cleaned in a 10 wt% Na_2CO_3 solution at 60 °C for 20 min, ensuring a thoroughly prepared surface for optimal welding outcomes.

The 5083 Al-Mg alloy plates were joined using a FSW machine to form a longitudinal butt joint. A conventional taper-threaded (right-handed) tool composed of H13 tool steel (52–55 HRC) was utilized as the rotating tool, set at a tilt angle of 2° in the anticlockwise direction, and subjected to an axial force of 11 kN. Welding was performed at three specific tool rotational speeds, i.e., 600, 700, and 800 RPM while maintaining a consistent traverse speed of 180 mm min⁻¹. The schematic diagram presenting the FSW process applied to the 5083 Al-Mg alloy plates is illustrated in figure 1.

To investigate the intermetallic compounds formed during the tool-stirring process between the plates, X-ray diffraction (XRD) analysis was carried out with an x-ray diffractometer (Equinox 2000, Thermo Fisher Scientific, USA). The system was equipped with a Ni filter and a solid-state detector, employing $\text{CuK}\alpha$ radiation ($\lambda = 1.5418 \text{ \AA}$) across a 2θ range of 10–110°.

Tensile test specimens were prepared in accordance with the ASTM E8/E8M standard. The tests were conducted at room temperature using a universal testing machine (LFM 300 kN, Walter Plus Baji, Switzerland) with a crosshead speed of $1.2 \times 10^5 \text{ mm min}^{-1}$. After tensile testing, the fractured samples were examined using a scanning electron microscope (SEM, FEI Inspect S50, USA) to analyze the fracture characteristics. Hardness testing was also performed to evaluate the hardness of the BM, TMAZ, and WZ zones. A Vickers hardness tester (Shimadzu Model HMV, Japan) was employed, using a diamond indenter with a 500 g load applied for 8 s. The final hardness values for each sample were calculated as the average of five individual readings.

The electrochemical behavior of FSW-welded 5083 Al-Mg alloy samples was comprehensively analyzed in a 3.5 wt% NaCl solution. Ground and polished samples with dimensions of 10 × 10 × 10 mm were cold-mounted in epoxy resin, exposing a 10 × 10 mm surface area to the electrolyte. Corrosion kinetics, mechanisms, and polarization behavior were examined using open circuit potential (OCP), potentiodynamic polarization (PDP), and electrochemical impedance spectroscopy (EIS) techniques. All measurements were conducted at $30 \pm 1 \text{ °C}$ using a potentiostat (Gamry Interface 1000, USA) in a three-electrode cell configuration. The system comprised an Ag/AgCl (with a saturated KCl electrolyte) reference electrode, a graphite rod serving as the counter electrode, and the FSW-welded 5083 Al-Mg alloy sample functioning as the working electrode.

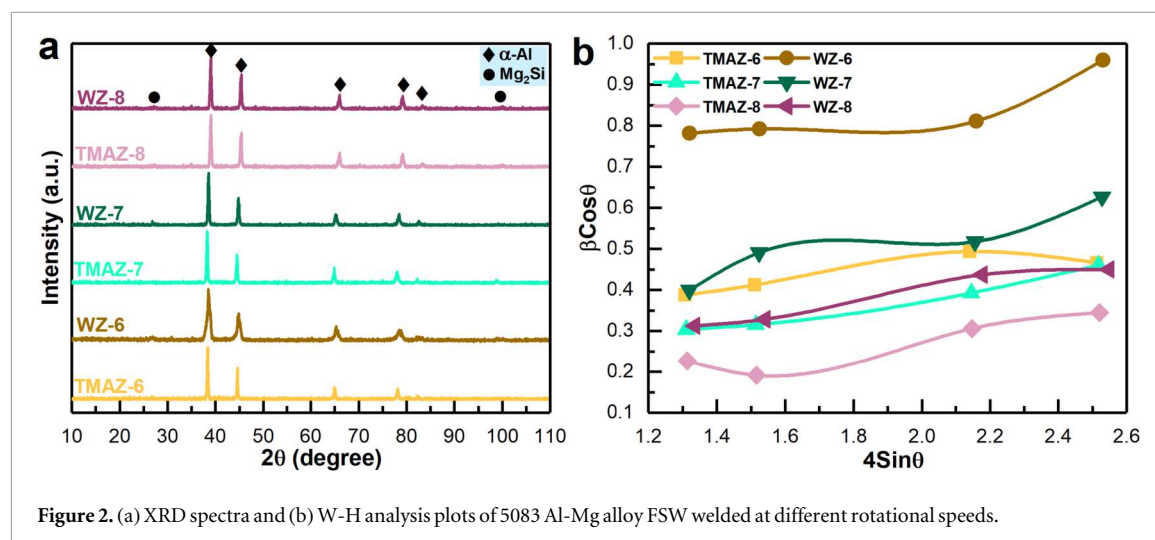


Figure 2. (a) XRD spectra and (b) W-H analysis plots of 5083 Al-Mg alloy FSW welded at different rotational speeds.

To evaluate the corrosion potential of the samples, OCP measurements were performed for 1 h to allow for stabilization. PDP analysis was conducted at a scan rate of 2.5 mV s^{-1} , ranging from an initial potential of -0.5 V to a final potential of 1 V (relative to OCP). EIS was carried out by applying a $\pm 5 \text{ mV}$ AC potential amplitude relative to OCP, covering a frequency range from 100 kHz to 0.01 Hz . Following the electrochemical tests, the corroded samples were analyzed using scanning electron microscopy (SEM) and energy-dispersive x-ray spectroscopy (EDX) to characterize the corrosion products formed on the surface during the analysis.

To further assess the long-term anti-corrosion properties of FSW-welded 5083 Al-Mg alloy plates, weight loss analysis was conducted following ASTM G-31/NACE TM 0169 standards. Samples measuring $10 \times 10 \times 10 \text{ mm}$ were wet-cut and ground using SiC papers up to P1000 grit. The initial weights of the samples were recorded prior to immersion in the solution. Each sample was suspended in a 20 ml cm^{-2} $3.5 \text{ wt\% NaCl}$ solution, as specified by the standard, within a container using a thread. The samples were periodically removed from the electrolyte every four days, cleaned with concentrated nitric acid, rinsed in water, and dried before recording their final weights. This procedure was repeated for a total duration of 60 days, with samples being re-immersed in the solution after each interval.

3. Results and discussion

3.1. Phase analysis

The XRD analysis provides profound insights into the structural characteristics of FSW-welded 5083 Al-Mg alloy samples, emphasizing the influence of rotational speed on grain size, residual stress distribution, and intermetallic compound formation. Figure 2 showcases the XRD spectra and Williamson–Hall plots, elucidating the variations in crystalline phases, grain size, and residual stresses across samples processed at rotational speeds of 600 RPM, 700 RPM, and 800 RPM. At 600 RPM (figure 2(a)), the broad diffraction peaks reflect significant residual stresses within the TMAZ and fine grains within the WZ zone. These residual stresses and grain refinement are primarily a result of the reduced axial force and low heat input ($\sim 410^\circ\text{C}$) which remains below the recrystallization temperature thereby limiting grain growth at this rotational speed compared to the 700 RPM and 800 RPM, which limit stress relaxation in the TMAZ. Additionally, low-intensity diffraction peaks at angles of 27° and 96° (JCPDS Card No. 35–0773) indicate the presence of Mg_2Si intermetallic compounds in the WZ zone. Although these compounds influence the mechanical properties of the weld, their limited formation at this speed leads to only a modest enhancement of the material's characteristics. At 700 RPM, the XRD diffraction peaks become narrower compared to those observed at 600 RPM, indicative of improved stress relief and enhanced dynamic recrystallization. The increased heat input ($\sim 430^\circ\text{C}$) and mechanical stirring during welding foster the development of finer grains within the WZ zone. This processing condition also encourages the formation of a greater quantity of Mg_2Si intermetallic compounds than at 600 RPM, albeit less than at 800 RPM. These intermetallic compounds, characterized by their distinctive XRD peaks at 26° and 98° , significantly enhance the weld's mechanical properties, contributing to increased strength and hardness. At 800 RPM, the diffraction peaks retain their narrow profile, signifying extensive stress relief in the TMAZ due to the high heat input ($\sim 470^\circ\text{C}$), which exceeds the recrystallization threshold. This elevated thermal condition results in slower cooling rates, facilitating the formation of uniformly distributed fine grains in the WZ. Furthermore, the higher temperature promotes the abundant

formation of Mg_2Si intermetallic compounds, confirmed by their characteristic XRD peaks at 26° and 98° [26]. The synergistic effect of refined grain structure and the uniform dispersion of intermetallic compounds ensures optimal mechanical properties for the weld at this rotational speed. Consequently, the 800 RPM sample demonstrates superior grain refinement and enhanced structural integrity compared to samples welded at lower rotational speeds, showcasing its exceptional performance characteristics.

The Williamson–Hall plot is an essential analytical method for quantifying residual stresses and micro-strain in materials, leveraging XRD data. In figure 2(b), at a rotational speed of 600 RPM, the TMAZ sample is characterized by elevated residual stresses. These high stress levels result from the rapid cooling and the axial force exerted by the tool during the welding process. The high cooling rate limits the time available for stress relaxation, leading to significant stress retention. This is clearly reflected in the XRD spectra, where pronounced peak broadening directly linked to residual stress levels is observed in the WZ sample. The combination of rapid cooling and insufficient stress relief at 600 RPM contributes to the high residual stress levels. Conversely, in the WZ zone is characterized by elevated residual stresses, as indicated by the pronounced broadening of diffraction peaks in the XRD analysis. The direct correlation between peak broadening and residual stress underscores the impact of rapid cooling at this lower rotational speed. The WZ zone experiences a high cooling rate due to the low heat input, leading to the development of a fine-grained microstructure and the formation of intermetallic compounds. This is evident from the steeper slope (figure 2(b)) of the 600 RPM curve compared to those at other rotational speeds, indicating higher residual stress levels. In contrast, at 700 RPM, the residual stress levels are markedly reduced compared to those at 600 RPM. The intermediate heat input at this speed promotes a slower cooling rate, allowing for more effective stress relief. This slower cooling supports the development of a more uniform grain structure in the WZ zone, which helps mitigate stress concentration. As a result, the XRD spectra for the 700 RPM sample display narrower peaks, indicating lower residual stresses. This balanced cooling rate and enhanced grain uniformity result in a more optimized microstructural configuration, reducing residual stress accumulation in the WZ sample [27]. In contrast, at a rotational speed of 700 RPM, the residual stress levels within the WZ zone are markedly lower, as reflected by the narrower and less broad peaks of the stress profile than 600 RPM. This reduction in residual stress is attributed to a more uniform microstructure, finer grain distribution, and a slower cooling rate compared to that at 600 RPM. At a rotational speed of 800 RPM, the elevated heat input relative to 600 and 700 RPM results in a slower cooling rate, which plays a pivotal role in enhancing residual stress relief. The extended cooling period allows for the effective dissipation of stresses, fostering the development of a highly uniform fine-grained microstructure within the WZ zone. This optimized thermal environment facilitates substantial stress relaxation, yielding significantly reduced residual stress levels at 800 RPM, as confirmed by narrower peaks in the XRD spectra. These findings highlight the favorable interplay of thermal dynamics and microstructural evolution at higher rotational speeds. In figure 2(b), the lower curve corresponds to the WZ sample at an 800 RPM rotational speed, which demonstrates reduced residual stress levels compared to the other samples [28]. This reduction is attributed to the significantly slower cooling rate at 800 RPM, which promotes the development of a more uniform fine-grained microstructure and enhances stress relaxation within the WZ sample. Therefore, it can be concluded that the residual stress levels at 800 RPM are lower than those at both 600 RPM and 700 RPM rotational speeds.

3.2. Mechanical properties

The tensile testing of FSW welded 5083 Al-Mg alloy samples was performed in accordance with the instructions outlined in the ASTM Standard E8/E8M. Figure 3(a) illustrates the stress–strain curves of BM 5083 Al-Mg alloy and FSW welded 5083 Al-Mg alloy welded at different rotational speeds of 600 RPM, 700 RPM, and 800 RPM. Figure 3(b) shows the samples after performing tensile testing or breakage, and highlighted portion show the exact locations from where the samples broke. The sample which was welded at 800 RPM breaks from the BM zone and the other 2 samples which were welded at 600 RPM and 700 RPM break from the FSW zone which clearly illustrates that the strength of the 800 RPM weld (342.48 MPa) is greater than 700 RPM and 600 RPM weld. At 600 RPM (335.47 MPa) and 700 RPM (340.33 MPa) welded samples, and it is obvious that the welds were weakly joined and are not safe in terms of mechanical properties [29]. Tensile strength of 800 RPM weld (342.48 MPa) also near to the strength of 5083 Al-Mg alloy metal (355.23 MPa). 800 RPM weld has high strength because at this rotational speed more heat generate in WZ zone and due to slow cooling more uniform fine grains form in it as compared to 700 RPM and 600 RPM weld.

Table 2 also illustrate that tensile strength of sample welded at 800 RPM rotational speed is higher than that of samples welded at 600 RPM and 700 RPM rotational speeds. During welding, the residual stresses in the 600 RPM generate higher than the 700 RPM and 800 RPM, so that's the other reason why the 600 RPM welded sample breaks from the welded zone. The elongation values alongside a tensile strength provide insights into the weld joints' ductility 800 RPM shows the maximum elongation of 12%, which indicates that the FSW process maintains the material's ability to undergo plastic deformation, vital for energy absorption and reducing

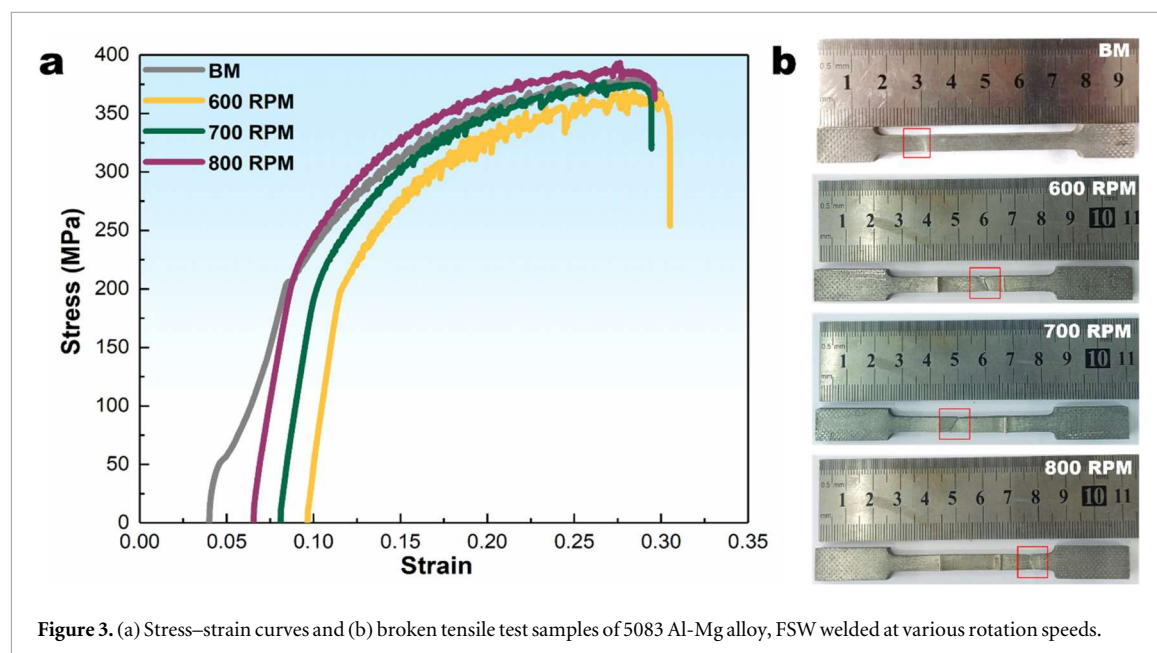


Figure 3. (a) Stress–strain curves and (b) broken tensile test samples of 5083 Al-Mg alloy, FSW welded at various rotation speeds.

Table 2. Tensile properties of FSW welded 5083 Al-Mg alloy at various rotational speeds.

Rotational speed (RPM)	Tensile strength (MPa)	Yield strength (MPa)	Elastic modulus (GPa)	Elongation (%)	Resilience (J.m^{-3})	Joint efficiency (%)
BM	355.23 ± 2.56	186.56 ± 1.50	69.81 ± 1.69	12.7 ± 0.74	9.7 ± 0.12	—
600	335.47 ± 1.96	163.89 ± 1.33	64.53 ± 1.66	10 ± 0.10	11.6 ± 0.14	94.36 ± 0.89
700	340.33 ± 2.53	167.11 ± 1.94	66.67 ± 1.34	11 ± 0.11	10.7 ± 0.22	95.77 ± 0.86
800	342.48 ± 1.11	169.91 ± 1.98	67.52 ± 0.55	12 ± 0.12	9.7 ± 0.23	96.33 ± 0.91

the risk of premature failure [30]. The elastic modulus values obtained at 600 RPM, 700 RPM, and 800 RPM were 64.53, 66.67 and 67.52 GPa, respectively, suggesting welded joints possess a comparable level of rigidity to the base material. This indicates the effective preservation of structural integrity and mechanical properties of the aluminum-magnesium alloy across all three rotational speeds. In table 2, the joint efficiency of the weld at 800 RPM rotational speed (96.33%) is exceeds that of the weld at 600 RPM and 700 RPM rotational speed (94.36% and 95.77%) respectively. This is why the 800 RPM tensile sample break from BM zone, whereas 600 RPM and 700 RPM tensile samples break from WZ zone due to lower joint efficiency.

Fractography analysis serves as a valuable method for examining the fractured surfaces of tensile samples. Its primary purpose lies in identifying the mechanism of sample failure. Figure 4(a), representing a tensile sample subjected to a rotational speed of 600 RPM, the fracture initiates within the WZ zone. Subsequent Fractographic examination reveals the presence of voids and inclusions on the fracture surface, resulting in reduced strength of material and failure within the WZ. As we know that voids and inclusions badly effect the material strength, the observable presence of voids and inclusions in the 600 RPM sample contribute to its lower mechanical properties, as also shown in the corresponding tensile graph [31].

Figure 4(b), illustrating a tensile sample exposed to a rotational speed of 700 RPM, the fracture once again initiates from the WZ zone. Fracture surface analysis mirrors the findings from the 600 RPM sample, described by the presence of voids and inclusions. These undesirable features, consistent with the 600 RPM sample, validate the WZ as the site of failure. Additionally, the presence of a dimple structure on the fracture surface points to the material's ductile behavior [32]. In the context of the 800 RPM tensile sample, fracture occurs within the BM zone rather than the WZ zone. Fractographic examination of this sample, as depicted in figure 4(c), reveals a distinct presence of dimple structures. This distinct structure signifies the material's ductile nature, supported by its elevated strength characteristics. Minimal presence of voids and inclusions in figure 4(c) further contribute to its superior strength. This trend is clearly evident in the corresponding tensile graph, where the 800 RPM sample exhibits the highest tensile strength (342.48 MPa) in comparison to the other tensile samples at 600 RPM and 700 RPM (335.47 and 340.33 MPa respectively). The examination of Vickers hardness provides insightful observations into the mechanical characteristics of our test specimen as shown in the figure 5.

The findings demonstrate distinct hardness variations in the WZ zone when subjected to rotational speeds of 600, 700, and 800 RPM. These hardness values surpass those of the BM and the TMAZ, indicating the

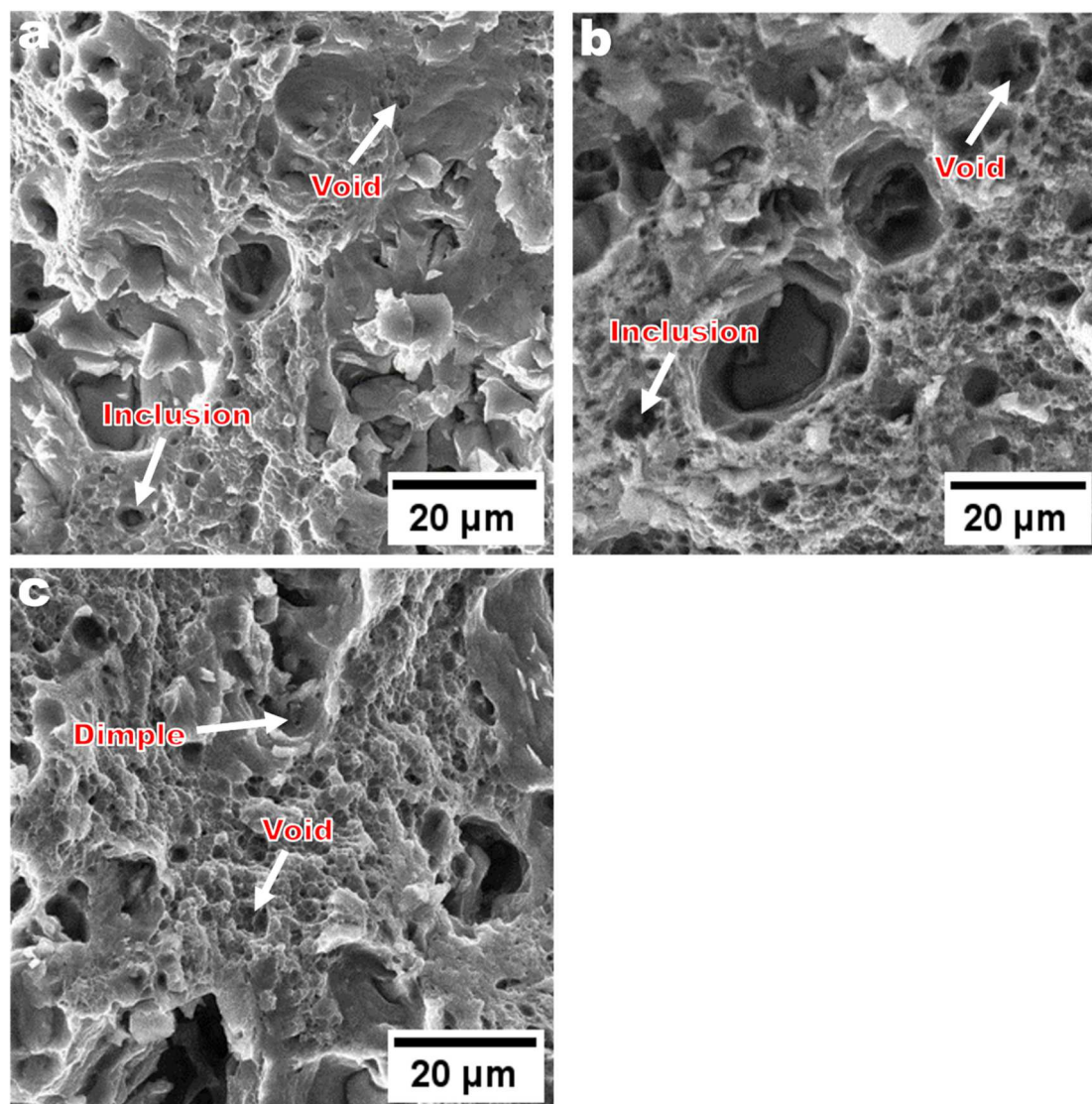


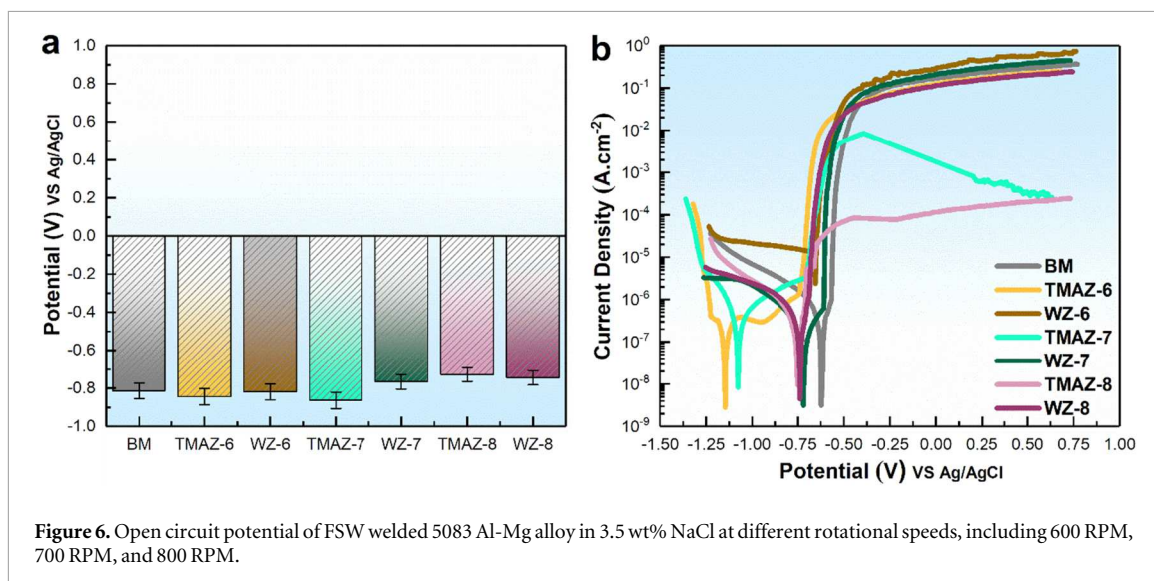
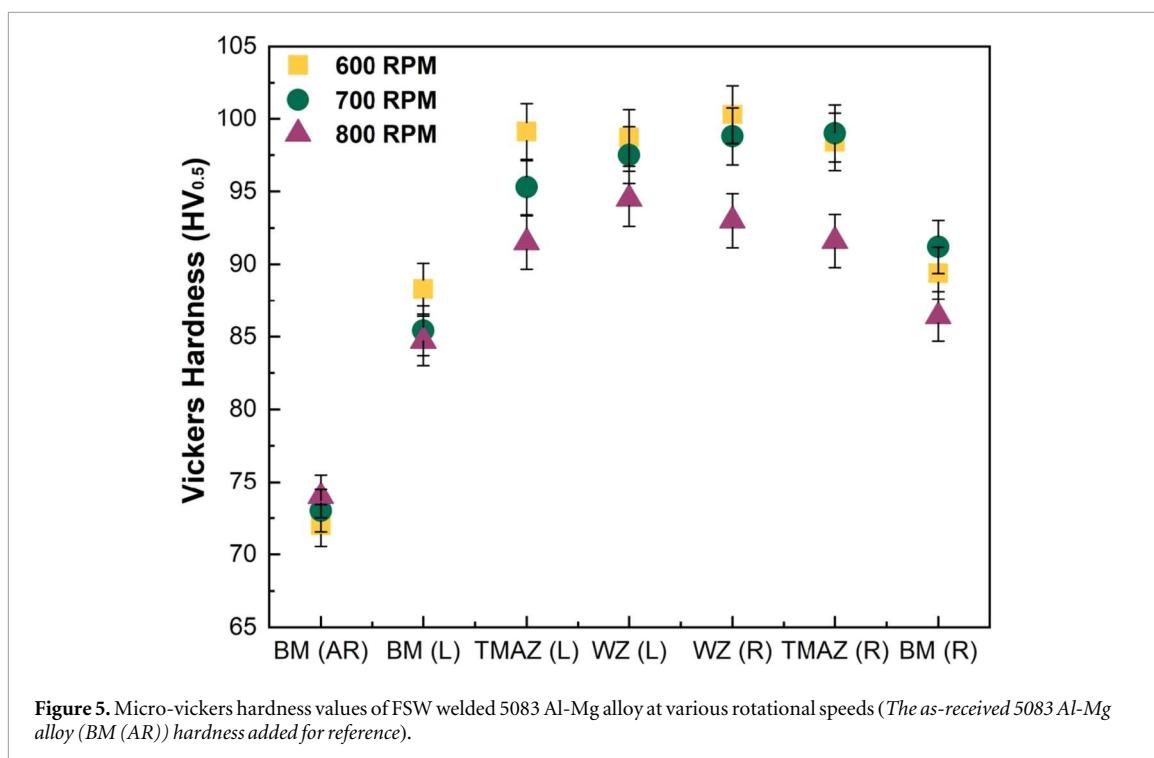
Figure 4. SEM fractographs of broken tensile tested 5083 Al-Mg alloy samples, FSW welded at various rotational speeds; (a) 600 RPM, (b) 700 RPM, and (c) 800 RPM.

capacity for enhanced properties post-application of the FSW technique. In figure 5 at a rotational speed of 600 RPM, the elevated hardness of the WZ zone (100.3 HV) is attributed to a relatively lower temperature (410 °C) reached during welding. This limited cooling time results in a higher hardness value due to the preservation of the material's microstructure. Higher rotational speeds, such as 700 RPM, result in a temperature increase to 430 °C, leading to extended cooling periods and a slightly reduced hardness of 97.5 HV compared to 600 RPM. However, excessive speed at 800 RPM raises the temperature to 470 °C, slowing down cooling and resulting in a lower hardness of 93 HV in the WZ zone, emphasizing the critical relationship between temperature and hardness values. These findings highlight the significance of process parameters in achieving desired mechanical properties [33].

3.3. Electrochemical properties

Open Circuit Potential (OCP) measures the electrochemical activity of a material in a given environment, which is unique for every element or an alloy. A more positive OCP indicates a lesser tendency of a material to react with its surrounding environment. OCP of TMAZ, WZ and BM of FSW welded 5083 Al-Mg alloy at three different rotational speeds (600, 700 and 800 RPM) was measured in 3.5 wt% NaCl for 3600 s is shown in figure 6(a). TMAZ of 800 RPM exhibited the most noble potential (−0.728 V), on the other hand, the most active potential was observed for TMAZ of 700 RPM (−0.863 V).

If we consider a comparative study of the galvanic behavior of all WZ zones at different rotational speed then it is clear that in case of 600 RPM, the WZ and TMAZ is more active as compared to BM zone, indicating their preferential dissolution [34]. In 700 RPM WZ has noble potential but TMAZ is active than BM zone.



Whereas, in 800 RPM both WZ and TMAZ exhibit a more positive potential than BM zone, suggesting that no corrosion would occur in the welded part. Moreover, the potential difference between TMAZ and WZ of 800 RPM is small, thus reducing the possibility of the formation of galvanic cell between both zones.

The polarization curves FSW welded 5083 Al-Mg alloy at 600, 700, and 800 RPM in 3.5 wt% NaCl has been shown in figure 6(b). The polarization measurements were done after an hour of OCP stabilization in the same solution. In 3.5 wt% NaCl solution, corrosive ions readily migrate to the oxide film Al_2O_3 , leading to the gradual dissolution of the aluminum alloy [35]. The aluminum matrix commonly behaves as an anode and dissolution takes place as reaction (1) and the second-phase particles becomes the cathode. In reaction (2) Hydroxide ions form the solution react with Al^{+3} ions to form a porous film of $\text{Al}(\text{OH})_3$ by the reaction (3).



In the solution, containing Cl^- ions compete with OH^- to adsorb on the surface of the oxide film. OH^- in the oxide film is gradually replaced, and with increasing corrosion time, the oxide film gradually becomes

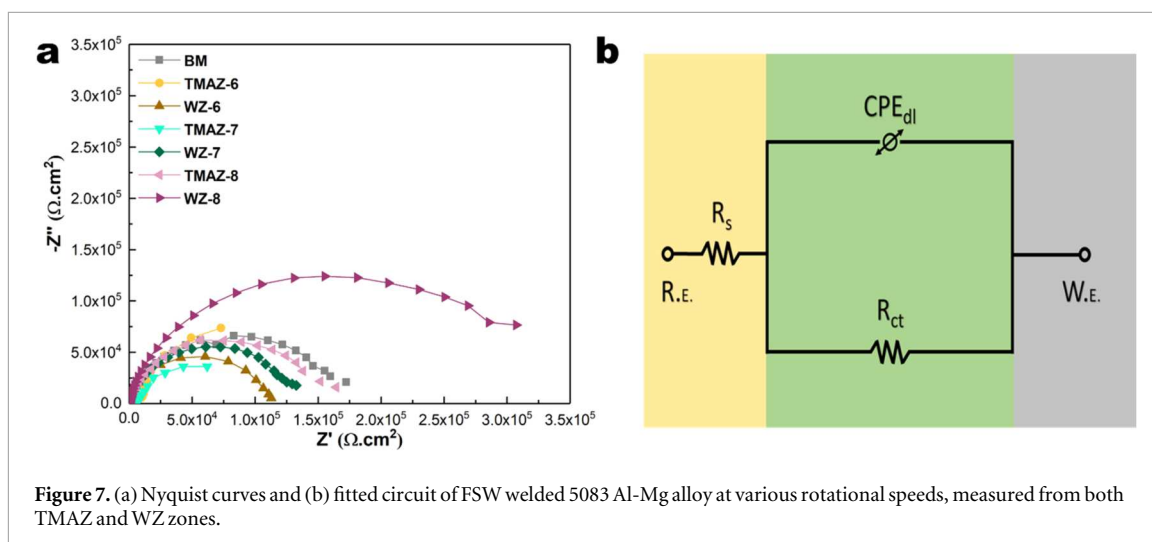


Figure 7. (a) Nyquist curves and (b) fitted circuit of FSW welded 5083 Al-Mg alloy at various rotational speeds, measured from both TMAZ and WZ zones.

Table 3. PDP test results of FSW welded 5083 Al-Mg alloy samples at various rotational speeds.

Sample ID	β_a (mV/decade)	β_c (mV/decade)	i_{corr} ($\mu\text{A} \cdot \text{cm}^{-2}$)	E_{corr} (mV)	Corrosion Rate (mpy)
BM	99.2	29.2	0.00087	−735	72
TMAZ-6	10.5	66.2	0.013	−1150	0.5
WZ-6	98.2	10.0	141	−746	61
TMAZ-7	41.5	16.2	0.051	−1080	0.2
WZ-7	99.9	89.3	0.9	−720	0.1
TMAZ-8	19.9	26.3	0.034	−746	0.03
WZ-8	117	13.7	0.09	−739	0.01

sparse and breaks forming AlCl_3 through reaction (4).



Values from Tafel fitting of polarization curves is illustrated in table 3.

The anodic polarization of the curves in all rotational speeds show a passivation behavior which can be attributed to the formation of passive oxide or hydroxide layer over the surface of sample. Figure 7 shows the Nyquist curves and the corresponding fitted circuit of FSW welded 5083 Al-Mg alloy in 3.5 wt% NaCl solution. All the Nyquist plots show single capacitive loop and the impedance value appeared at low frequency depicts the charge transfer resistance of the sample during the test. Greater R_{ct} value means higher resistance offered by the sample against its dissolution [36]. The equivalent circuit model in figure 7(b) simulate the overall electrical behavior of electrochemical cell formed between the reference (R.E) and working electrodes (W.E). R_s is the solution resistance, R_{ct} is charge transfer resistance and CPE_{dl} is the constant phase element of the double layer formed on the surface of the specimen. EIS test results are provided in table 4.

However, 800 RPM shows different trend, in which R_{ct} presented by the WZ zone is comparatively higher than BM. Also, the TMAZ of 800 RPM exhibits nearly same corrosion resistance as of the BM zone.

After conducting electrochemical test, we analyse all samples, using SEM to determine what compounds are formed on the surface of sample during electrochemical analysis. In figure 8(a) BM zone of FSW welded 5083 Al-Mg alloy exhibits surface cracks on surface and AlCl_3 compound formed which can lead to corrosion in form of pits or small holes. A high corrosion rate is measure at 600 RPM rotational speed (61 mpy). In figure 8(b) we can see that intensity of chloride ions high that contribute in formation of AlCl_3 ions, resulting in severe corrosion (0.5 mpy) due to voids approximately 100 μm in size. These voids typically occur in regions of insufficient heat input or improper tool penetration in the TMAZ at 600 RPM [37]. Similarly, in 600 RPM WZ figure 8(c) intensity of chloride ions high which causes pitting corrosion on the surface of weld. At 700 RPM rotational speed corrosion rate decreases significantly (0.1 mpy) as compared to 600 and 800 RPM rotational speed. In figure 8(d), the TMAZ at 700 RPM demonstrates a lower intensity of chloride ions, resulting in minor voids of approximately 50 μm with irregular shapes on the alloy surface. In the WZ at 700 RPM, as shown in figure 8(e), AlCl_3 formation is still observed, causing pitting corrosion and small holes of about 20 μm . In figure 8(f) at 800 RPM TMAZ although fewer elements react to the alloy surface and low intensity of Cl^- ions

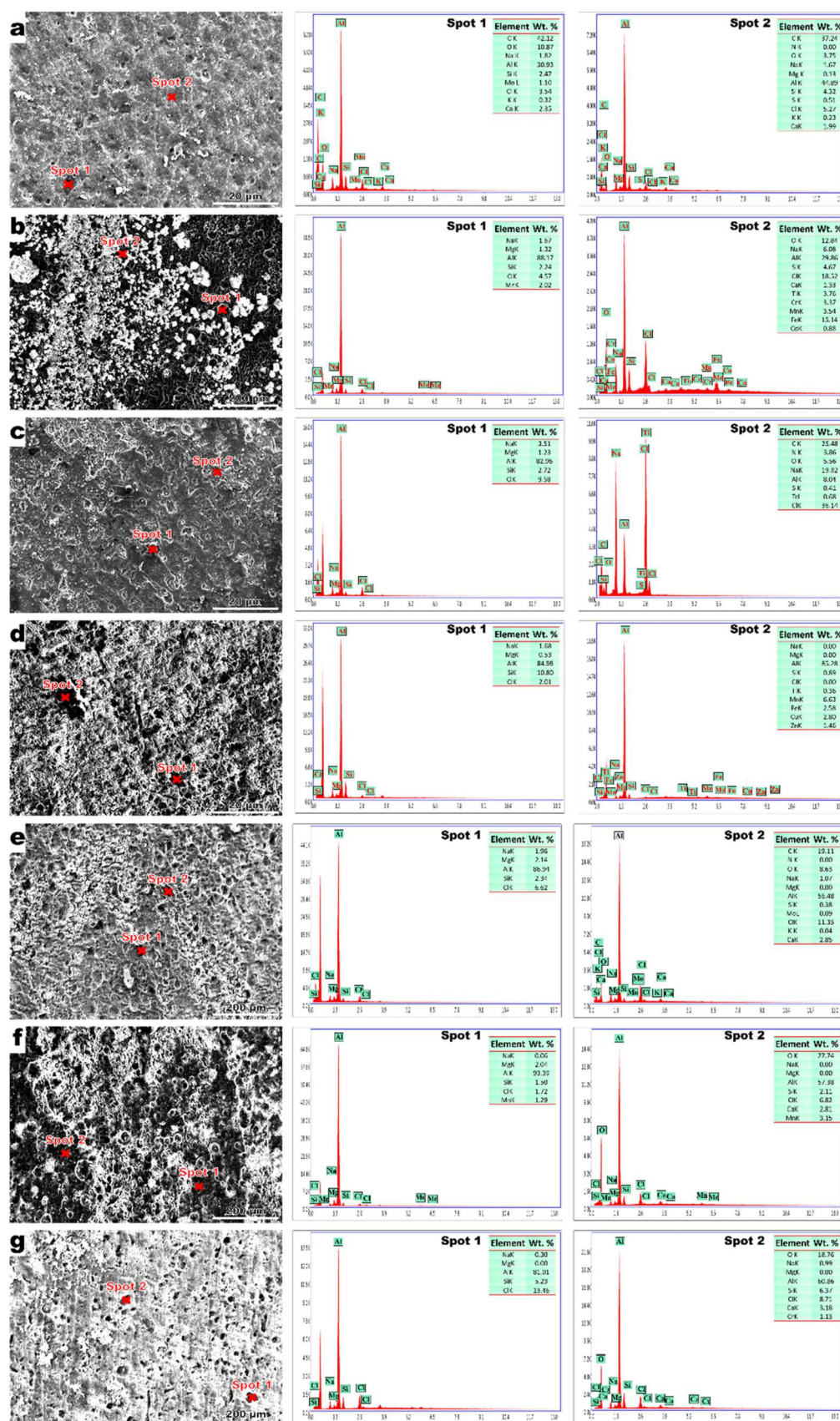


Figure 8. SEM micrographs and corresponding EDS spectra of FSW welded 5083 Al-Mg alloy samples, after electrochemical analysis; (a) BM, (b) TMAZ-6, (c) WZ-6, (d) TMAZ-7, (e) WZ-7, (f) TMAZ-8, and (g) WZ-8 samples.

lead to formation of AlCl_3 ions which can corrode (0.03 mpy) TMAZ and cause pitting corrosion. At 800 RPM WZ in figure 8(g) some cracks and holes appear due to presence of AlCl_3 compound that can decrease the corrosion rate (0.01 mpy) [38].

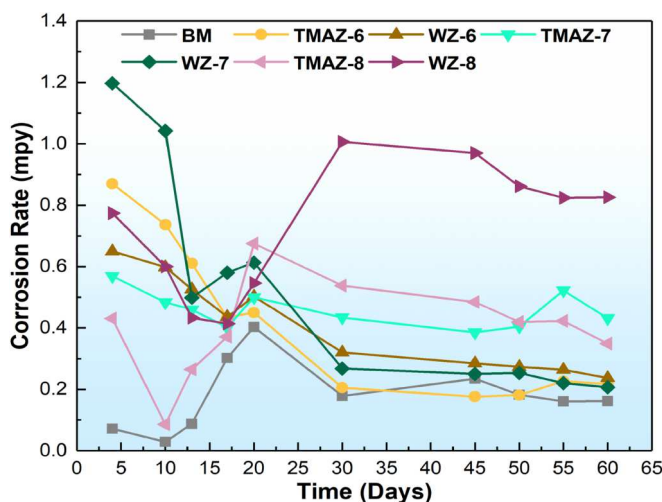


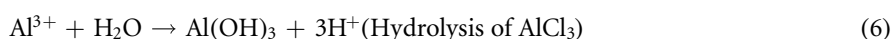
Figure 9. Plot of corrosion rates of FSW welded 5083 Al-Mg alloy samples as a function of immersion time.

Table 4. EIS test results of FSW welded 5083 Al-Mg alloy samples at various rotational speeds.

Sample IDs	R_s ($\Omega \text{ cm}^2$)	CPE_{dl} ($\mu\text{F cm}^{-2}$)	n	R_{ct} ($k\Omega \text{ cm}^2$)
BM	4.42	06.597	0.890	212.301
TMAZ-6	5.19	11.886	0.860	72.816
WZ-6	4.49	08.026	0.902	112.832
TMAZ-7	6.78	17.665	0.797	61.900
WZ-7	5.77	06.394	0.894	170.072
TMAZ-8	7.02	100.07	0.319	173.440
WZ-8	3.23	05.072	0.924	344.099

It is generally recognized that aluminum alloys are vulnerable to chloride (Cl^-) corrosion, especially in the marine atmosphere. The oxide film formed near the intermetallic compounds is thinner than in other areas. Hence, preferential corrosion will tend to occur where this film is damaged by chloride attack, leading to formation of pits. The pitting corrosion propagates according to the following electrochemical reactions (6)–(7).

Oxidation at the anode (inside a pit):



Reduction at the cathode (outside the pit cavity):



Previous studies have suggested that for electron transfer the pits would appear near the intermetallic compounds. The formation of Al^{3+} ions from the oxidation reaction takes place at the bottom of a pit leading to migration of Cl^- ions toward this position in order to balance the positive charge on the Al^{3+} ions thus resulting in AlCl_3 formation inside each pit. Hydrolysis of AlCl_3 then leads to formation of HCl and Al^{3+} with the HCl at the root of the pit causing acidification up to a pH value < 3 , which further accelerates pit propagation [39]. The Al^{3+} ions concentrated inside the pit will diffuse towards the pit opening and form $\text{Al}(\text{OH})_3$ corrosion product via the reduction reaction at the lateral surfaces near the top of the pit.

Corrosion rate of each sample was calculated using the formula.

$$\text{corrosion rate} = \frac{\Delta m}{A \times t \times D} \quad (9)$$

Where ' Δm ' is the weight loss in grams, ' A ' is total surface area of the sample in mm^2 , ' t ' is immersion time in days and ' D ' is the density of alloy in g cm^{-3} [40].

Figure 9 shows the weight loss trend of FSW welded 5083 Al-Mg alloy measured for 60 days in 3.5 wt% NaCl. The TMAZ zone 600 and 700 RPM rotational speeds exhibit a similar pattern in which, initially the corrosion rate is high, i.e., 0.87 mpy, 0.65 mpy and it decreases gradually to 0.23 mpy and 0.20 mpy respectively as the immersion time increases. This trend can be attributed to formation of stable oxide layer that acts as a

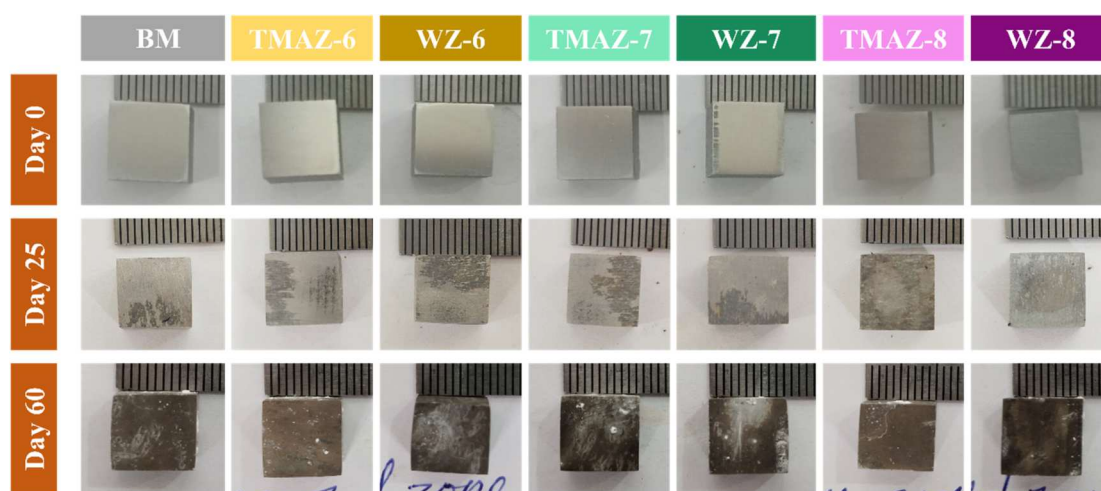


Figure 10. Images of FSW welded 5083 Al-Mg alloy samples before and after 25 and 60 days immersion in 3.5 wt% NaCl.

physical barrier in the dissolution process. However, no notable drop-in corrosion rate with the passage of time was observed in TMAZ of 800 RPM.

For WZ zones, 700 and 800 RPM shows a similar behavior, where corrosion rate decreases from 0.77 mpy and 0.43 mpy to 0.43 mpy and 0.08 mpy respectively in the initial 10–12 days of immersion [41]. After that a rise in corrosion rate appeared and the final values for 700 and 800 RPM were 0.35 and 0.82 mpy. The initial reduction in dissolution behavior can be related to the formation of passive oxide layer, which is also observable from figure 10. The action of Cl^- ions caused the corrosion rate to increase again. The BM zone has the lowest corrosion rate among all the zones and was least affected by the increasing immersion time [42, 43].

4. Conclusions

The influence of tool rotational speed on the microstructure, mechanical properties, and electrochemical behavior of FSW welded 5083 Al-Mg alloy was comprehensively investigated. The key findings from this study are as follows:

- Welding at a rotational speed of 800 RPM resulted in the lowest levels of residual stress and the highest tensile strength of 342.48 MPa, surpassing the performance of samples welded at 600 RPM and 700 RPM.
- SEM analysis of fractured tensile samples welded at 600 RPM and 700 RPM, revealed the presence of voids and inclusions. These microstructural defects were linked to the diminished mechanical properties observed at these lower rotational speeds when compared to the 800 RPM sample.
- Electrochemical assessments demonstrated that the corrosion rates of the WZ and TMAZ samples welded at 800 RPM were 0.01 mpy and 0.03 mpy, respectively. These values represented the lowest corrosion rates among the three rotational speeds tested, indicating improved corrosion resistance at 800 RPM.
- Welding at a rotational speed of 800 RPM provided a microstructure with abundant formation of Mg_2Si intermetallic compounds. These intermetallic compounds refined the microstructure, improved hardness, and enhanced joint integrity without introducing brittleness, thereby contributing to the superior overall performance of the weld, enhanced mechanical strength, and superior electrochemical stability, establishing it as the most advantageous parameter among those investigated.

Data availability statement

The data cannot be made publicly available upon publication because they are not available in a format that is sufficiently accessible or reusable by other researchers. The data that support the findings of this study are available upon reasonable request from the authors.

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